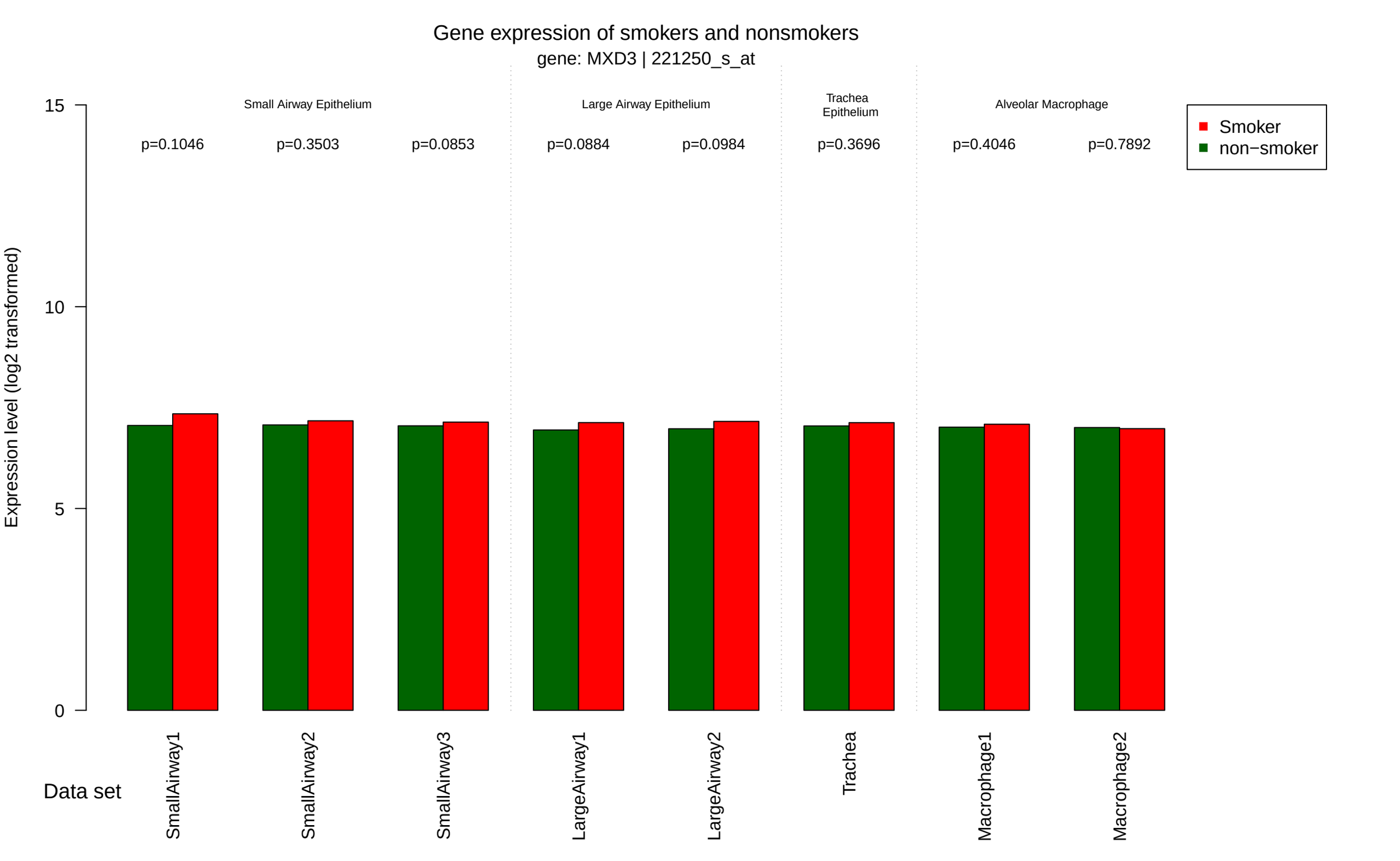
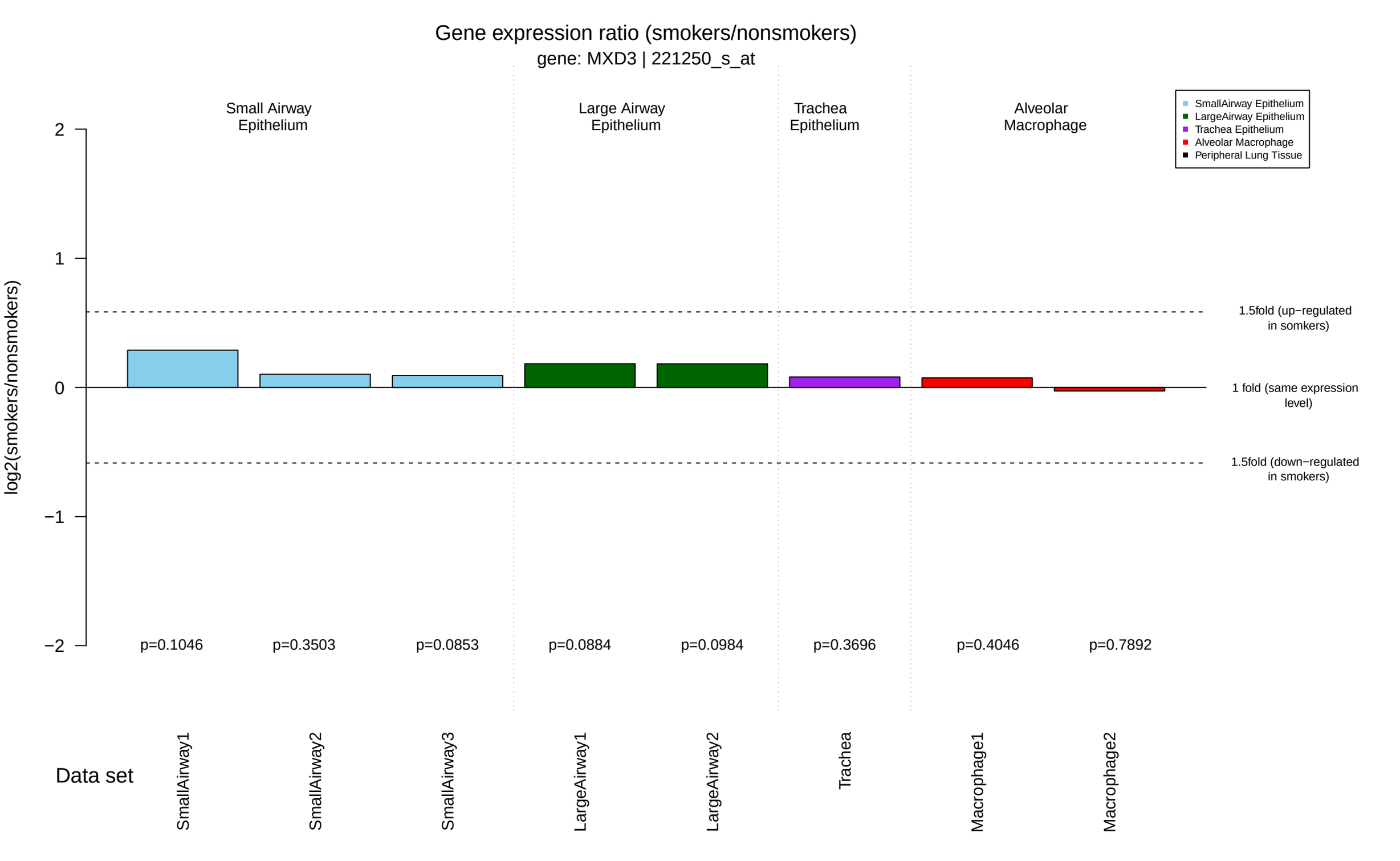
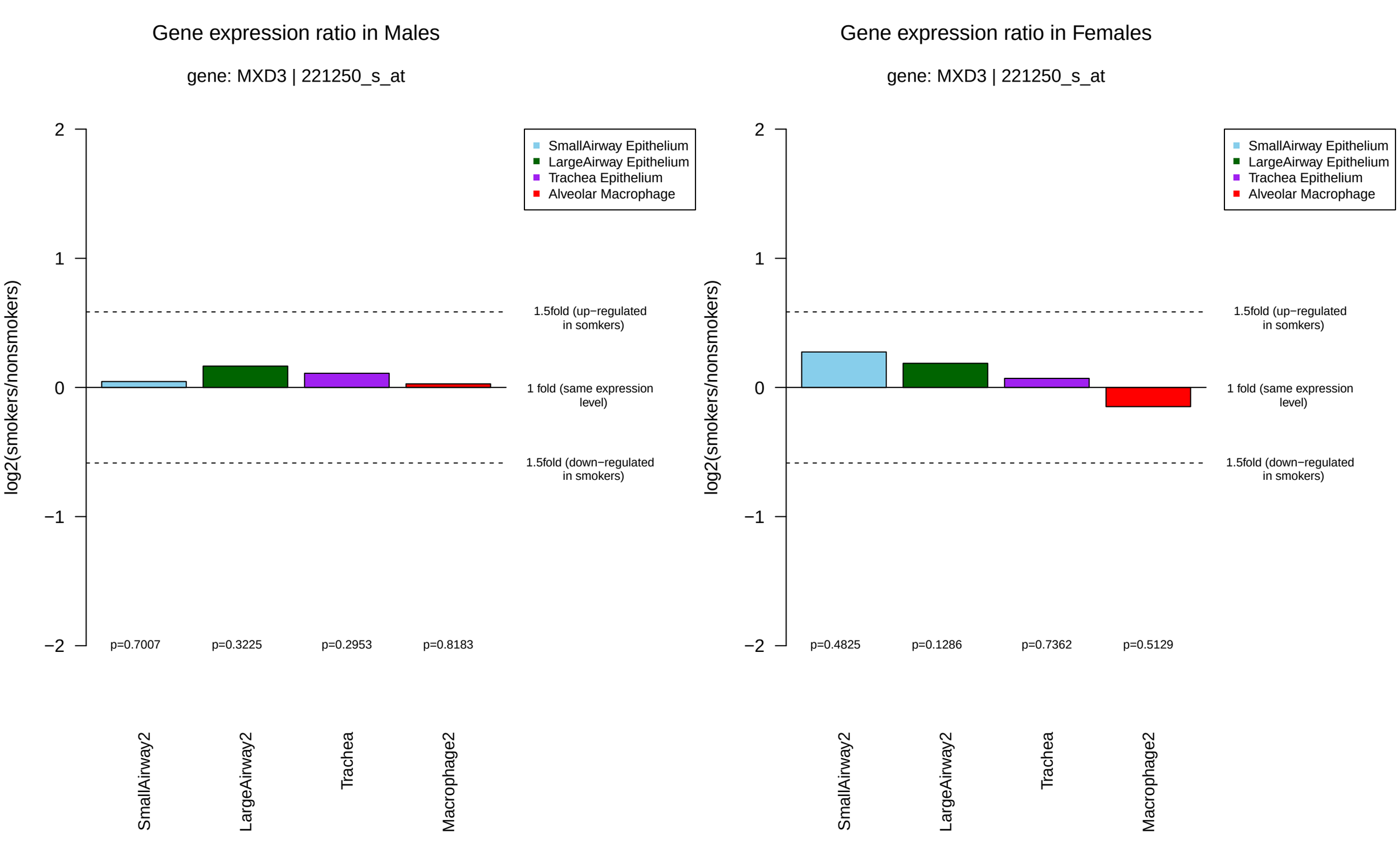
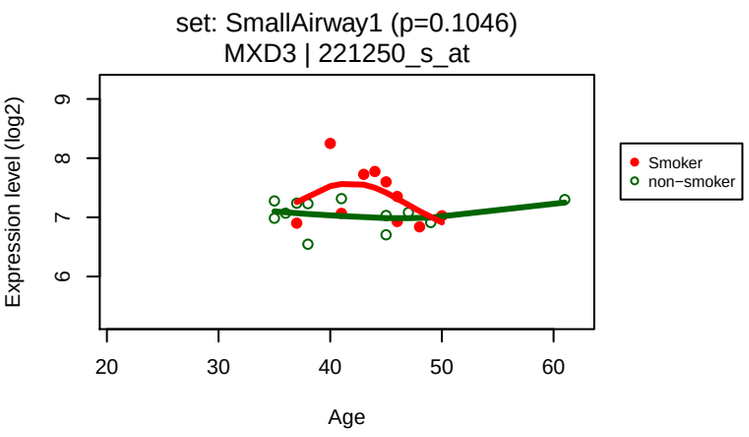
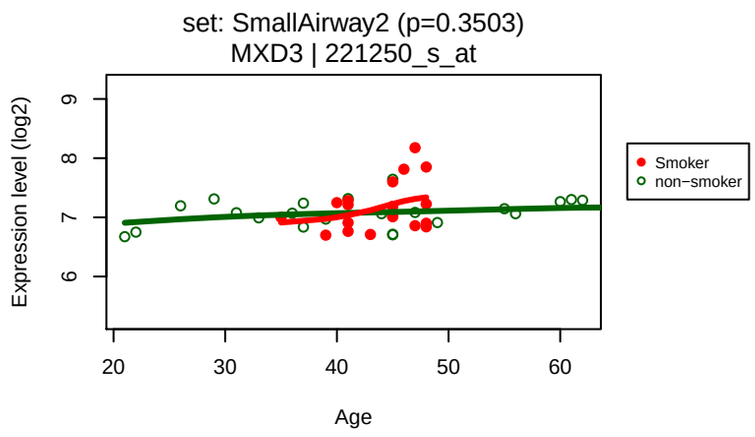
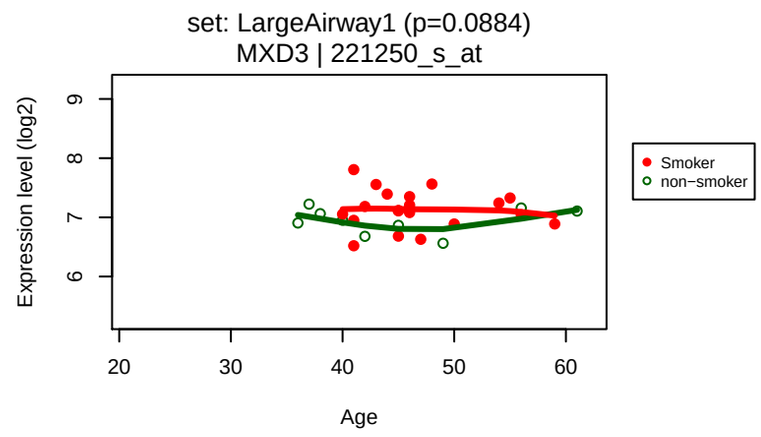

Gene expression of smokers and nonsmokers
gene: MXD3 | 221250_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1046
p=0.3503
p=0.0853
p=0.0884
p=0.0984
p=0.3696
p=0.4046
p=0.7892
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

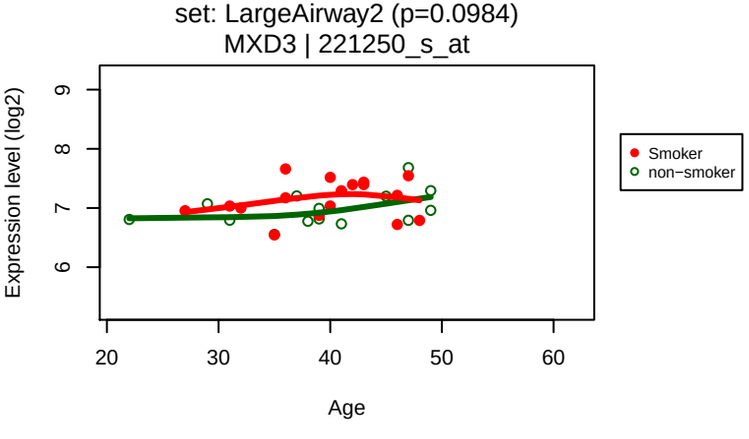

Gene expression ratio (smokers/nonsmokers)
gene: MXD3 | 221250_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1046
p=0.3503
p=0.0853
p=0.0884
p=0.0984
p=0.3696
p=0.4046
p=0.7892
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

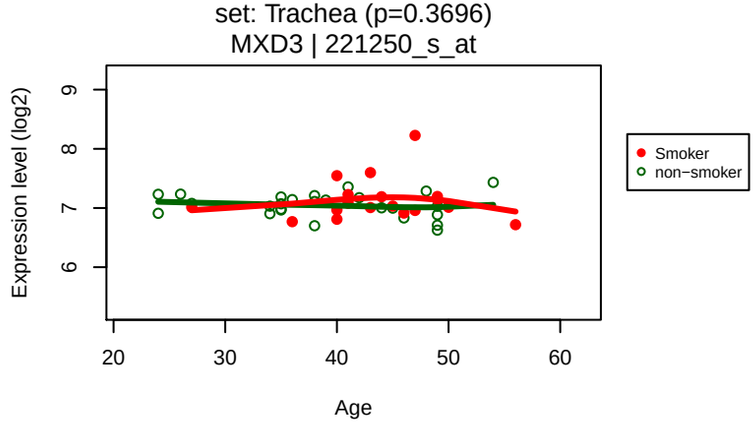

Gene expression ratio in Males
gene: MXD3 | 221250_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7007
p=0.3225
p=0.2953
p=0.8183
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MXD3 | 221250_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4825
p=0.1286
p=0.7362
p=0.5129
SmallAirway2
LargeAirway2
Trachea
Macrophage2

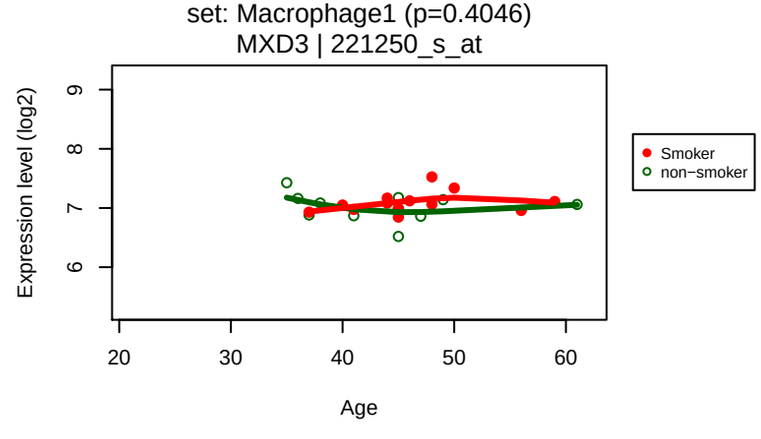

set: SmallAirway1 (p=0.1046)
MXD3 | 221250_s_at
Expression level (log2)
Age
Smoker
non−smoker

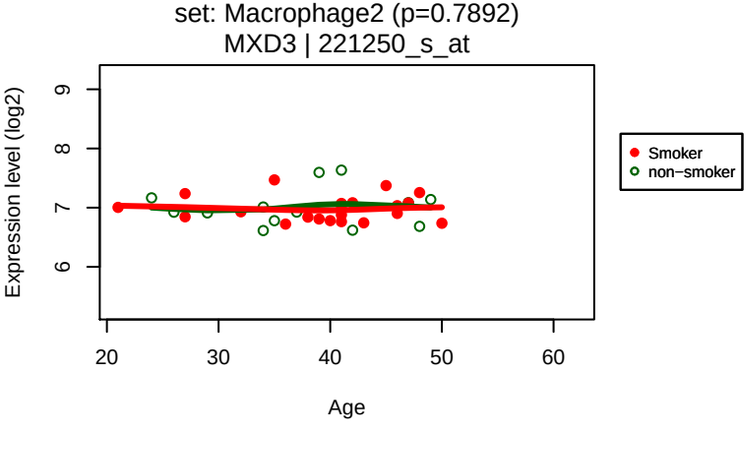

set: SmallAirway2 (p=0.3503)
MXD3 | 221250_s_at
Expression level (log2)
Age
Smoker
non−smoker

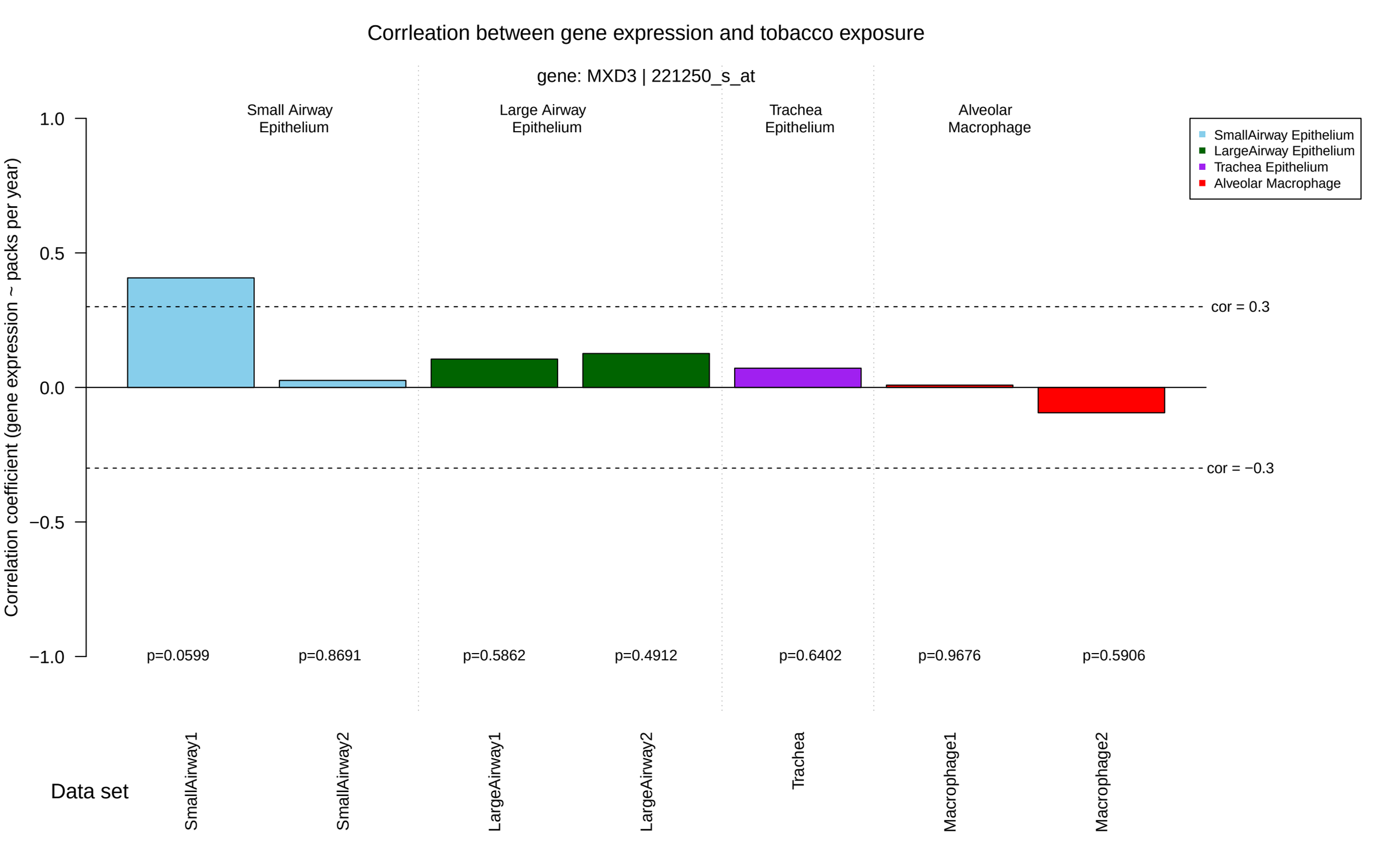

set: LargeAirway1 (p=0.0884)
MXD3 | 221250_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0984)
MXD3 | 221250_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3696)
MXD3 | 221250_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4046)
MXD3 | 221250_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7892)
MXD3 | 221250_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MXD3 | 221250_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0599
p=0.8691
p=0.5862
p=0.4912
p=0.6402
p=0.9676
p=0.5906
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2